

2017 MJC H2 Chemistry Prelim Suggested Answers

- 1(a) (i) An acid is a substance that can donate a proton (H^+) to another substance/proton donor.

A base is a substance that can accept a proton (H^+) from another substance/proton acceptor.

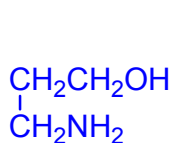
- (ii) *Arrhenius theory* of acids and bases applies to aqueous system only.
or
Arrhenius theory of acids and bases does not apply to acids/bases that do not dissociate H^+/OH^- from itself.

(b) (i)
$$\frac{[H^+]_{water}}{[H^+]_{methanol}} = \frac{\sqrt{10^{-2.45}}}{\sqrt{10^{-7.50}}} = 331$$

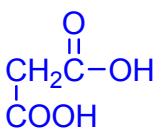
- (ii) Water is a more polar solvent than methanol. (hence the both H^+ and conjugates bases can better stabilized through the formation of stronger ion-dipole interations).
- (iii) $CH_2(CN)COOH$, can form more extensive hydrogen bonding with water due to its additional $-CN$ group .

- (c) (i) In dimethylamine, there are more electron-donating $-CH_3$ groups that increases the electron density of the lone pair on N atom which increases the availability of the lone pair on N to accept H^+ .
- (ii) Trimethylamine have 3 bulky methyl groups that causes steric hindrance. Hence it reduces the availability of lone pair of electron on N atom to accept H^+ .

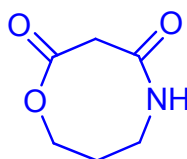
- (d) (i)



A



B



D

- (ii) Step I: acidic hydrolysis
Step II: Nucleophilic acyl substitution
- (iii) Step I: dilute sulfuric acid, heat
Step III: dilute sulfuric acid, heat followed by $LiAlH_4$, dry ether, rtp

2(a) Zn obtained from daily feed = $0.7 \times 180 \times 3 / 100 = 3.78$ mg

(It meets the recommended quantity for daily intake)

(b) (i) I: oxidation

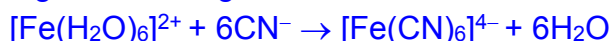
II: acid-metal displacement

(ii) Cation in A: $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$

Gas B is H_2 .

$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ has very high charge density, it polarises the electron cloud of the surrounding water molecules, weakens and breaks the O-H bond, which results in the release of a proton.

(iii) Ligand exchange



(iv) 1. Fe^{2+} has partially-filled d-orbitals.

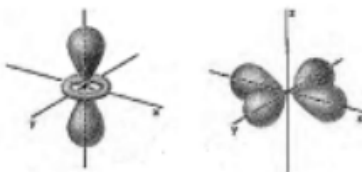
In $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$, the d orbitals are split into two groups due to the ability of the ligands to split them.

A lower energy d electron undergoes d-d transition and is promoted to a higher energy d orbital. During the transition, the d electron absorbs the red wavelength of light from the visible region of the electromagnetic spectrum and the remaining wavelengths not absorbed appear as the green colour.

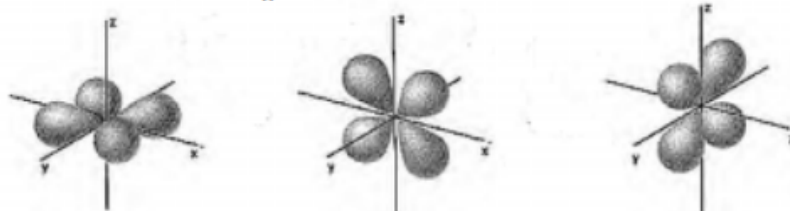
2. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$

3. $[\text{FeF}_5(\text{H}_2\text{O})]^{2-}(\text{aq})$ is colourless because the energy absorbed for d-d transition is out of the range of the visible spectrum.

(v) correct upper orbital diagram



correct lower orbital diagram



Any one from each category

- (c) (i) Fe has more protons and hence has greater nuclear charge than Ca. But there is only slight increase in screening effect as electrons are added to the inner 3d orbitals that provide shielding effect for the 4s electrons from Ca to Fe.

Hence, Fe has greater effective nuclear charge and has increased electrostatic attraction between the nucleus and the valence 4s electrons compared to Ca. Therefore, atomic radius of iron is less than calcium.

- (ii) The relative atomic mass of Fe is larger than that of Ca and the atomic radius of Fe (0.126) is smaller than that of Ca (0.197 nm) thus the densities of Fe and Cu is greater than that of Ca.

- (d) (i) MgCl_2 solution: pH = 6.5 PCl_5 solution: pH = 2.0



- (ii) Element **Z** belongs to Group 13. $\Rightarrow$ **Z** is Al.
Element **F** could be element **Z**.

- 3(a) **Y** is the plot for NOCl . This is because of its permanent dipole–permanent dipole interaction between its molecules which is most significant, hence it deviates most from ideal gas .

(b) (i)
$$n = \frac{5.80 \times 10^6 \times 5.0 \times 10^{-3}}{8.31 \times (250 + 273)} = 6.67$$

(ii)
$$K_p = \frac{(P_{\text{NO}})^2 (P_{\text{Cl}_2})}{(P_{\text{NOCl}})^2} \quad \text{Pa}$$

- (iii) Let x be the change in number of moles of gas.

$$\text{Total number of moles of gases at equilibrium} = 5 - 2x + 2x + x$$

$$x = 1.673$$

At equilibrium,

$$\text{No. of moles of NOCl} = 1.65$$

$$\text{No. of moles of NO} = 3.35$$

$$\text{No. of moles of Cl}_2 = 1.67$$

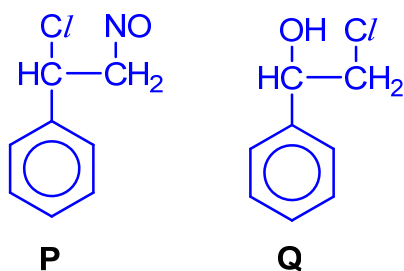
$$K_p = 5.95 \times 10^6 \text{ Pa}$$

- (c) (i) $K_c = 1.40$

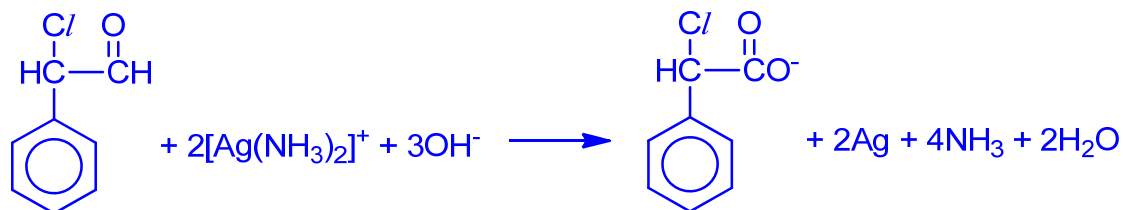
- (ii) When T increases, K_c decreases .
- (iii) Since K_c decreases, the amount of reactants will increase and products will decrease .

By *Le Chatelier's* Principle, when temperature is raised and equilibrium position shifts left , the backward reaction is endothermic so as to absorb heat . Therefore the forward reaction is exothermic .

(d) (i)



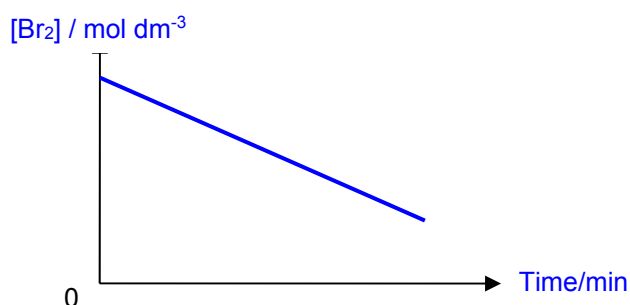
(ii) Tollens' reagent



Accept alternative reagents

4(a) (i) Order of reaction w.r.t to CH_3COCH_3 is 1 .
Order of reaction w.r.t to H^+ is 1

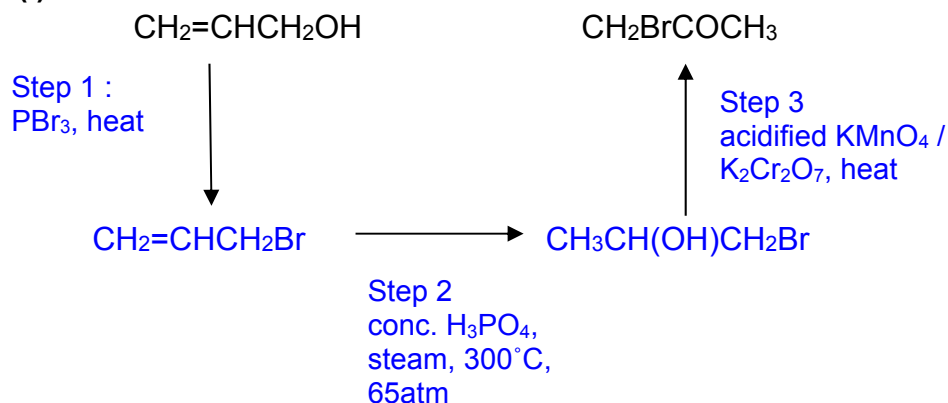
(ii)



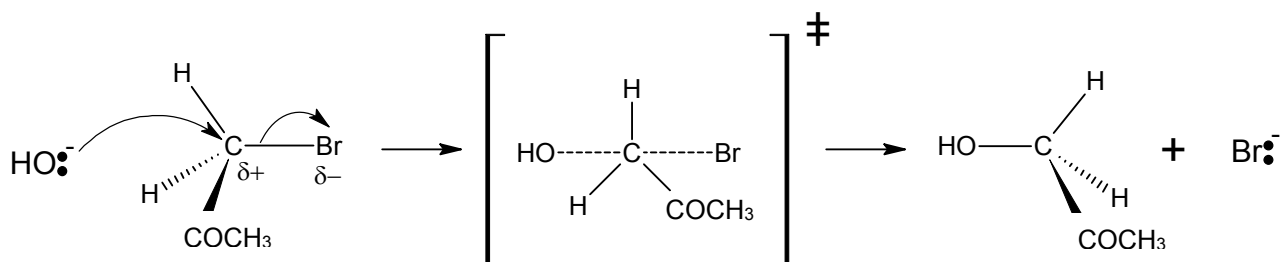
(iii) $\text{Rate} = k[\text{CH}_3\text{COCH}_3][\text{H}^+]$

(iv) $k = 3.80 \times 10^{-3} \text{ mol}^{-1} \text{ dm}^3 \text{ min}^{-1}$

(b) (i)



- (ii) Add aqueous Br_2 , r.t.p to the sample.
 If the orange Br_2 decolourises, $\text{CH}_2=\text{CHCH}_2\text{OH}$ is present, reaction is not complete.

(c) (i) Nucleophilic substitution $\text{S}_\text{N}2$ 

- (ii) $\text{CH}_3\text{CH}_2\text{COBr}$ has a faster rate of hydrolysis.

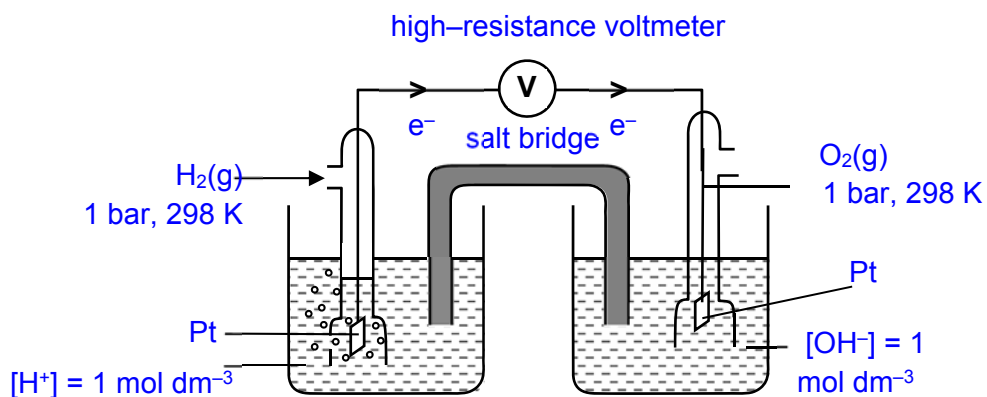
The carbonyl C atom in $\text{CH}_3\text{CH}_2\text{COBr}$ is bonded to two electronegative atoms O and Br which makes the carbonyl C atom highly electron-deficient and is very susceptible to nucleophilic substitution.

5(a) (i) Anode: $\text{H}_2 + \text{CO}_3^{2-} \longrightarrow \text{H}_2\text{O} + \text{CO}_2 + 2\text{e}^-$

Cathode: $\frac{1}{2} \text{O}_2 + \text{CO}_2 + 2\text{e}^- \longrightarrow \text{CO}_3^{2-}$

$\text{H}_2 + \frac{1}{2} \text{O}_2 \longrightarrow \text{H}_2\text{O}$

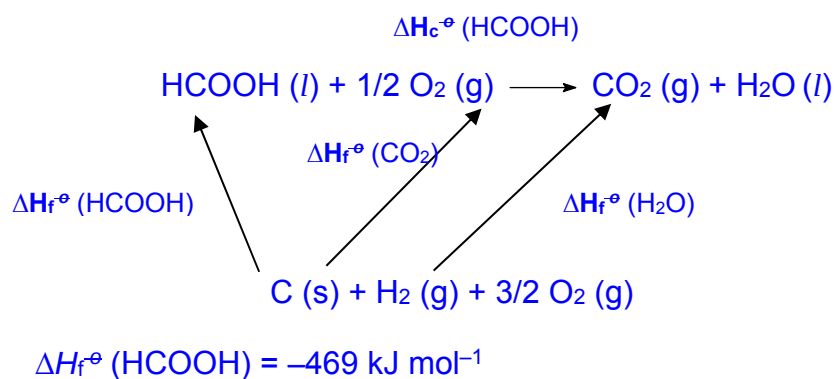
(ii)



(b) more negative

(c) (i) $H_f^\ominus$ of formic acid is the energy change when one mole of the formic acid is formed from its elements at standard conditions of 298K and 1 bar.

(ii)



(iii) $\Delta H_{\text{rxn}}^\ominus = (-150) + (-286) - (-246) = -190 \text{ kJ mol}^{-1}$